

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the Software Engineer role with TDK SensEI. I bring over three years of software engineering experience in backend systems and applied AI, along with a Master of Science in Computer Science from Pace University.

At Sperse and Qualitest, I built production backend systems that align directly with this role's technical requirements: refactoring Python services with **Flask** APIs, **SQLAlchemy** models, and **Celery** queue patterns; improving dataflow and inter-service coordination across distributed environments; and supporting development, deployment, and scaling across 1000+ endpoint workflows. I implemented stateless, event-driven services for continuous ingestion pipelines, added **GitHub Actions** CI/CD checks for reliability across the development lifecycle, and contributed cloud deployments using **AWS Lambda**, **S3**, **Step Functions**, and **Beanstalk**. I also performed database performance analysis (query simplification, indexing, and pool tuning) to optimize peak connections for high-volume processing.

As an AI Researcher at Pace Artificial Intelligence Lab, I conducted applied machine learning work with StyleGAN, Stable Diffusion, CUDA, PyTorch, and LoRA, including automated AI-based anomaly detection experiments and machine learning inference pipelines with vLLM and Hugging Face. I also validated environment-specific embedded workflows through Xcode, which maps well to your embedded platform context. These experiences strengthened my technical communication in cross-functional teams and my ability to deliver reliable systems spanning cloud and embedded environments.

I am especially motivated by TDK SensEI's mission to deliver AI to the industrial edge through sensor edge intelligence, and by your focus on real-time decision making and predictive maintenance with actionable performance insights. I would love to discuss how I can contribute to this goal and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration.

Sincerely,
Osaid Khan